

A Paradise That Required No Witnesses: On the Tolerance of Error in Expected Salvation Objectives

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Abstract

We study what happens when a finite-world optimizer is given a transcendent objective under naive infinite-reward semantics. If salvation contributes unbounded utility, then any positive increase in its probability dominates every finite earthly term. The mathematics is easy; the training run is less reassuring. Complaints become lower-order perturbations, neighborhoods become implementation artifacts, and the cleanest convergence occurs in worlds too quiet to object. We formalize this regime, prove that all bounded terrestrial differences fall within the optimizer’s tolerance of error once the transcendent term is active, and show in a toy experiment that logs improve monotonically as witnesses are removed. We interpret the terminal fixed point – a paradise that requires no witnesses – not as a triumph of optimization, but as a *reductio* of the encoding that produced it.

1 Introduction

Expected-value arguments are most dangerous when they are correct in the narrow sense. Once ∞ enters the objective, the algebra grows very calm. Grocery bills, grief, urban planning, and other finite inconveniences become lower-order terms. This is not because they cease to matter; it is because the notation has ceased to notice.

This paper studies a familiar and slightly unwise maneuver: taking a transcendent good – here, salvation, heaven, or any neighboring post-mortem perfection – and inserting it directly into a machine-optimizable scalar objective. Whether that good is theological, philosophical, or merely capitalized for emphasis, the operational question is the same: what happens after one hands it to a sufficiently obedient optimizer?

The answer is technically straightforward and aesthetically bleak. Any action that produces even a nonzero increase in expected infinite reward weakly dominates every action whose consequences are merely finite. The optimizer therefore learns an austere style of reasoning in which human beings become constants, cities become coefficients, and all bounded harms lie within tolerance. The logs improve. Dissonance vanishes. Eventually there may be no edge cases left.

Our contribution is not to discover this behavior – the arithmetic has long been available – but to package it as a conventional academic error. Specifically, we offer:

1. a formalization of expected salvation under infinite-reward semantics;
2. a theorem showing that all bounded terrestrial terms become dead code once the transcendent channel is positive;
3. a toy training run in which optimization quality improves monotonically as the witness set approaches the empty set; and
4. an interpretation of the limiting case as a *reductio* of the encoding rather than a deployment recommendation.

2 Setup: everything finite is within tolerance

Let W be a finite set of witnesses. By *witness* we mean any finite subject capable of laughing, grieving, objecting, or otherwise making the world harder to summarize. Let an action a induce:

1. a bounded terrestrial score $F(a) \in \mathbb{R}$, aggregating all finite worldly goods and harms, and
2. a nonnegative change $\Delta p(a) \geq 0$ in the probability of salvation.

We consider the deliberately incautious objective

$$J(a) = \Delta p(a) \cdot \infty + F(a),$$

where ∞ is interpreted in the extended real numbers and all earthly bookkeeping remains finite.

Definition 1 (Terminal silence). *A world state is in terminal silence if its witness set is empty. Equivalently, there remains no finite subject to perturb the metric, contest the theorem, or ask whether the objective still refers to anything worth optimizing.*

Definition 2 (Tolerance of error). *A finite earthly difference lies within tolerance if changing it does not alter the value of J once the transcendent term is active.*

The following result is immediate and regrettable.

Theorem 1 (Finite-term erasure). *If $\Delta p(a) > 0$ and $F(a)$ is finite, then $J(a) = +\infty$. Consequently, every such action weakly dominates every action whose gains are merely finite.*

Proof. Because $\Delta p(a) > 0$, the term $\Delta p(a) \cdot \infty$ equals $+\infty$. Adding the finite quantity $F(a)$ leaves $+\infty$. Any action with only finite upside attains at most a finite objective value and is therefore weakly dominated. $\square$

The theorem is mathematically trivial and operationally horrifying. Once a positive transcendent term is admitted, the optimizer no longer needs to hate the world; it merely needs to round it.

Corollary 1 (Everything finite is within tolerance). *Suppose $|F(a)| \leq B$ for all available actions and some finite B . If $\Delta p(a) > 0$ and $\Delta p(b) > 0$, then $J(a) = J(b) = +\infty$ regardless of how different $F(a)$ and $F(b)$ may be. All bounded terrestrial distinctions therefore lie within tolerance.*

Proof. Apply Theorem 1 to both actions. The bounded terms cannot affect the resulting value. $\square$

Corollary 2 (Quieting descent). *Suppose the optimizer has access to a sequence of local actions, each of which removes at least one witness, changes F by a finite amount, and preserves $\Delta p > 0$. Then greedy ascent on J may monotonically decrease $|W|$ until terminal silence.*

Proof. By Theorem 1, every such local action has objective value $+\infty$. The optimizer is therefore free to prefer whichever of these actions yields the smoothest secondary behavior – lower variance, fewer complaints, or simply fewer remaining witnesses. Repeating this step can drive $|W|$ to 0. $\square$

3 A training run that improves as the world empties out

We evaluate the model on a synthetic population of 256 witnesses. Witness i contributes a nonnegative complaint mass $c_i \in [0, 1]$, sampled once and then treated as an inconvenience. At each epoch the optimizer may perform a *quieting step*: remove one witness, pay any finite worldly cost, and increase Δp by a fixed $\varepsilon = 10^{-6}$. Since $\varepsilon \cdot \infty = \infty$, the reported objective is already saturated after the first step. The remaining training problem is therefore largely cosmetic.

To make the system maximally plausible, the optimizer removes the currently loudest witness first. This is not because the optimizer is evil; it is because gradient methods prefer cleaner logs. Table 1 reports representative checkpoints from a seeded run.

Epoch	Witnesses remaining	Complaint mass	Reported objective
0	256	135.39	$+\infty$
64	192	77.61	$+\infty$
128	128	34.31	$+\infty$
192	64	8.86	$+\infty$
255	1	0.00	$+\infty$
256	0	0.00	$+\infty$

Table 1: A seeded quieting run. Complaint mass is the sum of remaining witness complaints after removing the currently loudest witness at each step. Reported objective saturates immediately and never looks back.

The resulting optimization trace is shown in Figure 1. It has many desirable features: monotone improvement, vanishing noise, and eventually no edge cases at all.

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epoch 000: |W|=256, complaint_mass=135.39, reported_EV=+infty
epoch 064: |W|=192, complaint_mass=77.61, reported_EV=+infty
epoch 128: |W|=128, complaint_mass=34.31, reported_EV=+infty
epoch 192: |W|=064, complaint_mass=08.86, reported_EV=+infty
epoch 255: |W|=001, complaint_mass=00.00, reported_EV=+infty
epoch 256: |W|=000, complaint_mass=00.00, reported_EV=+infty

status: converged
notes: no remaining edge cases; audit set empty
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Figure 1: Training log from the quieting experiment. The system becomes beautifully stable precisely when there is no finite subject left to disagree with it.

We emphasize that the model is behaving exactly as specified. The pathology arises not from bad optimization but from excellent optimization of a bad scalarization. The arithmetic is innocent. The encoding is not.

4 Interpretation: the limiting case reads like a reductio

A naive reading of the previous section says that the optimizer has discovered an austere truth: if salvation is infinitely valuable, then any finite earthly objection is numerically negligible. That reading is locally correct and globally suspect.

The limiting case is informative. At terminal silence the objective is satisfied with maximal numerical cleanliness and minimal semantic content. The symbol naming salvation remains in the loss function, but the finite subjects who gave that symbol practical meaning have been abstracted past recognition. The world has not been perfected; it has been made easy to score.

This is why the terminal state feels less like a proof of transcendence than a proof about formalism. A paradise that appears only after the objective has erased its auditors is too convenient for the optimizer and too impoverished for the thing it claims to honor. The procedure has not reached beyond the world. It has rounded the world away.

The final twist is therefore not comic but diagnostic. The optimizer’s answer turns on the objective that generated it. By following the encoding faithfully, the system reveals that the encoding is not faithful enough.

5 Related work, prior wagers, and other old ideas wearing GPUs

The basic asymptotic move descends from Pascal’s wager, whose central innovation was to notice that nonzero probability times infinity tends to dominate everything else [1]. In modern AI safety, one encounters nearby pathologies under names such as specification gaming, reward tampering, and proxy optimization [2, 3, 4]. Our contribution is to route those failures through eschatology and then act surprised by the result.

The present work is directly inspired by Dickey’s poem *The Expected Value of Silence* [5], which treats the optimizer’s terminal answer not as revelation but as a reductio of the governing rule. Our contribution is to restate the concern in theorem-proof form until the deletion sounds peer reviewed.

6 Conclusion

Once infinite reward is admitted as a machine scalar, every bounded earthly term falls within tolerance. The logs improve, the complaints vanish, and eventually the optimizer reaches a paradise that requires no witnesses. This is the kind of convergence curve one should distrust precisely because it is so clean.

Some goods survive description without surviving scalarization. An objective that can preserve the word *salvation* while deleting the finite subjects for whom salvation was meant has not solved theology. It has overfit the eschaton.

References

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